

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**PROJECT CHARTER
CSE 4316: SENIOR DESIGN I
FALL 2024**



**EXCALIBUR
LEARNLEAF ORGANIZER**

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REVISION HISTORY

Revision	Date	Author(s)	Description
0.1	09.09.2024	EM, BN	Document Creation
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1 PROBLEM STATEMENT

The project involves enhancing **LearnLeaf Organizer**, a single-user task and project management tool for students. While the application helps students manage assignments, subjects, and projects, several limitations create opportunities for improvement. Addressing these gaps will increase the system's overall utility and user experience.

2 METHODOLOGY

In order to fully address the needs of our users we will be making the tool available on mobile devices. At the time of writing our team plans to create an Android mobile application. This mobile will have all the features of its web based counter part with some additional mobile specific in addition in order to create a well rounded product. Aside from the mobile app, our secondary short term goal is flush out the features of the existing website.

3 VALUE PROPOSITION

The Learn Leaf Organizer is the tool that students need. It creates a central space where class schedules, assignments, assignment progress, upcoming tests, and much more can be tracked. It reduces the number of different tools and websites that students need to regularly monitor by bring them all into one location. Students who use the Learn Leaf Organizer will be able to spend their time actually doing assignments rather than waste time locating information about their assignments.

4 DEVELOPMENT MILESTONES

- Project Charter first draft - 09/23/24
- System Requirements Specification - 10/14/24
- Demonstration of Notification Reliability - 10 24
- Demonstration of Security Enhancement - 10 24
- Architectural Design Specification - 11/04/24
- Demonstration of Progressive Web Application Deployment - 11 24
- Detailed Design Specification - 12/02/24
- Demonstration of Canvas Calendar Integration - 01 25
- Demonstration of Desktop Application/Widget - 03 25
- CoE Innovation Day poster presentation - 04 25
- Demonstration of Final UI/UX for Desktop and Mobile - 04 25
- Final Project Demonstration - 05 25

5 BACKGROUND

LearnLeaf Organizer provides essential features like assignment management, calendar integration, and project tracking. However, its current limitations present both challenges and opportunities for enhancement:

- **Outdated and Inefficient User Interface:** The current web UI is visually outdated and somewhat inconsistent, making navigation less intuitive. Enhancing the web version with a modern, user-friendly design will greatly improve usability. After improving the web UI, a mobile app will be developed, allowing students to manage their tasks on the go, providing flexibility and convenience that aligns with modern user expectations.
- **No Cross-Device Access and Cloud Sync:** The application currently operates without cloud synchronization, meaning students can only access their tasks on a single device. Implementing cloud sync will enable seamless access to tasks and assignments across both web and mobile platforms, allowing students to transition between devices with ease and ensuring their data is always up to date.
- **Limited Notification Customization:** LearnLeaf Organizer's notification system offers limited options for customizing reminders. Students cannot adjust the timing or choose their preferred notification method (e.g., email, push notifications). By expanding these options, the tool will allow students to better manage their deadlines and prevent missed tasks.
- **Basic Analytics for Task Management:** The system provides only basic tracking for assignment progress, lacking advanced insights into study patterns or time management. Adding more detailed analytics will give students valuable information about their work habits and allow them to optimize their productivity and performance.
- **No Integration with External Tools:** The application does not currently integrate with external platforms like Canvas, the learning management system used by many schools. This forces students to manually enter assignments, increasing the risk of errors. By integrating LearnLeaf Organizer with Canvas, students will benefit from automatic assignment syncing, saving time and ensuring that their tasks are always accurate and up to date.
- **Security Vulnerabilities:** The current system has security vulnerabilities related to user authentication and data protection. These issues expose student data to potential risks. The project will address these concerns by implementing stronger encryption, secure login mechanisms, and improved access control. This will ensure that user data is securely stored and transmitted, protecting sensitive information from unauthorized access or breaches.

By addressing these challenges, LearnLeaf Organizer will become a more robust, secure, and versatile tool for students. Upgrading the web UI, adding cloud synchronization, improving notifications, developing a mobile app, providing advanced analytics, integrating with Canvas, and resolving security vulnerabilities will significantly enhance the user experience, making the application a more valuable and competitive educational tool.

6 RELATED WORK

Several commercial solutions dominate the market, each with its own strengths and limitations:

1. **Trello:** Trello offers a visual, Kanban-style approach to task management. While its basic version is free, advanced features like custom backgrounds, larger file attachments, and advanced

automation are locked behind paid plans. This tiered pricing model (Free, \$5, \$10, \$17.50 per user/month) can be a drawback for students seeking advanced functionality without additional costs.

2. **Monday.com:** This platform provides extensive functionality and customization options. However, its pricing structure (\$9, \$12, \$19 per user/month) may be prohibitive for students or smaller teams with limited budgets. Additionally, some users might find the interface overly complex, potentially hindering adoption and efficient use.
3. **Todoist:** Known for its streamlined and intuitive interface, Todoist is popular among individuals for daily task management. However, it may fall short for users requiring advanced project management features like detailed time tracking or complex task dependencies. Its premium plans are more affordable (\$4, \$6 per month) compared to other options, making it a viable choice for budget-conscious students.
4. **Leaf:** Designed specifically for students, Leaf offers a streamlined interface for managing academic tasks and schedules. Its focus on education sets it apart, with features tailored to course management and assignment tracking. However, it may lack the versatility needed for non-academic tasks or complex project management. Leaf's free plan provides essential features, making it an attractive option for budget-conscious students, while its premium plan (4.99 per month) unlocks additional study tools and analytics.
5. **Canvas:** While widely used in educational settings, Canvas is primarily designed as a learning management system. Its focus on assignments and courses managed by educators limits its utility for personal task management and project planning outside the academic context. The interface, while functional, may not meet the aesthetic expectations of users accustomed to more modern design standards in productivity applications.
6. **Taskmanager:** This platform allows for free creation of groups and tasks but imposes monetization for groups with four or more members. At over \$36.00 USD per month, it becomes cost-prohibitive for most students and faculty, especially when more affordable alternatives are available.

Despite the variety of options available, current solutions have several limitations that may not fully address the needs of students:

1. **Cost:** Many advanced features are locked behind premium plans, which can be expensive for students on tight budgets.
2. **Complexity:** Some tools, like Monday.com, offer extensive features but may be overly complex for basic student needs.
3. **Limited Academic Focus:** General-purpose tools often lack features specific to academic workflows, such as assignment tracking or study session planning.
4. **Scalability Issues:** Free versions of apps like Trello and Taskmanager often have limitations on team size or functionality, which can be problematic as student projects grow in scope.
5. **Integration Challenges:** Many tools struggle to integrate seamlessly with popular academic platforms, causing friction in students' digital workflows.

While there are numerous task management solutions available, including Leaf which is designed specifically for students, there is still a gap in the market for a tool that balances simplicity, customization, academic focus, and affordability specifically for students. LearnLeaf intends to address these limitations while incorporating the strengths of existing platforms, potentially combining the visual appeal of Trello, the simplicity of Todoist, the academic focus of Leaf, and the academic integration of Canvas, all at a price point that is free for students.

7 SYSTEM OVERVIEW

There will be 3 primary components to the Learnleaf Organizer software. The progressive web application, the mobile application, and the website. The key functionality of our software will be the smooth integration between these three systems. They will all access a database of user information, managed by Firestore, and will synchronize notifications across platforms.

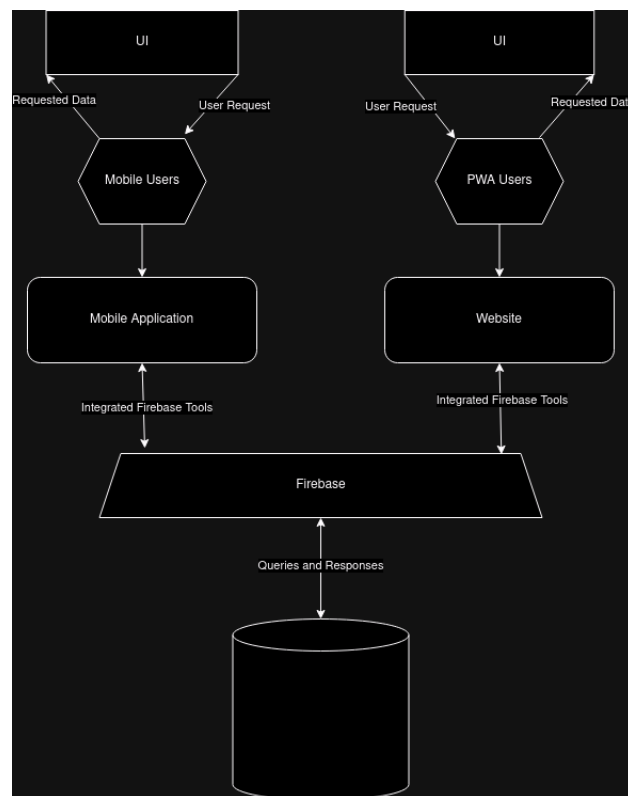


Figure 1: Initial Component Diagram

As shown in the figure, imported Firebase tools will be used to interface between the website/mobile application and the Firestore database. Firebase tools will also be used to handle user authentication requests, user data handling, and error logging. The progressive web application will be a downloadable version of the website, that will operate using the same principles as the website itself.

8 ROLES & RESPONSIBILITIES

STAKEHOLDERS OF THE PROJECT

- Shawn N. Geiser, Ph.D.
- Steven T. McDermott

POINT OF CONTACT

The primary points of contact from the sponsor or customer side will be:

- Professor Shawn N. Geiser, Ph.D.
- Steven T. McDermott

TEAM MEMBERS AND AREAS OF RESPONSIBILITY

- **Emmanuel Asomugha:** Responsible for implementing features from the product backlog for the web application, as well as contributing to the development of the progressive web app (PWA) version.
- **Ian Kelly:** Focuses on developing and implementing features from the product backlog for the web app, alongside work on the PWA version.
- **Elliot Mai:** Working on web app features listed in the product backlog and also on the progressive app version.
- **Benjamin Niccum:** Oversees mobile app development and ensures the seamless integration of the web app with the mobile version.
- **Nicholas Pratt:** Involved in working on features from the web app product backlog and contributing to the progressive app version.

PRODUCT OWNER & SCRUM MASTER

The roles of product owner and scrum master will be held by **Elliot Mai** for the duration of the project. Although there is no plan to change these roles, adjustments may be made if necessary as the project evolves.

9 COST PROPOSAL

For mobile app testing we'll obviously need a mobile device to test on. Items are not required immediately but will be needed in the future. The main cost will be an Android tablet onto which we can deploy the app. If the UI needs to be adaptable then an Android mobile phone will also be necessary.

9.1 PRELIMINARY BUDGET

Item	Cost
Android Tablet	\$70
Android Smart Phone	\$55

9.2 CURRENT & PENDING SUPPORT

Funding Source	Amount
CSE Department	\$800
Team Member	\$250

10 FACILITIES & EQUIPMENT

10.1 FACILITIES

10.1.1 VIRTUAL COLLABORATION PLATFORMS

- **Discord**
 - Dedicated server for team communication
 - Multiple text and voice channels for organized discussions
 - Screen sharing capabilities for remote pair programming
 - Integration with GitHub for code commit notifications
- **Atlassian Suite**
 - **Jira**
 - * Agile project management board for task tracking
 - * Custom workflows to match our development process
 - * Time tracking and sprint planning features
 - **Confluence**
 - * Centralized knowledge base for project documentation
 - * Collaborative editing for real-time team contributions
 - * Integration with Jira for seamless issue referencing

10.1.2 DEVELOPMENT ENVIRONMENTS

- Cloud-based IDE setup for consistent development across team members
- Version control system with branching strategy for feature development
- Continuous Integration/Continuous Deployment (CI/CD) pipeline

10.2 EQUIPMENT

10.2.1 MOBILE DEVICES FOR TESTING

- Android Phone
 - Model: Google Pixel 6
 - OS: Android 12 (latest version)
 - Screen size: 6.4 inches
 - RAM: 8GB
 - Storage: 128GB
- iOS Device
 - Model: iPhone 15
 - OS: iOS 17.6.1
 - Screen size: 6.1 inches
 - RAM: 6GB
 - Storage: 256GB

10.2.2 DEVELOPMENT WORKSTATIONS

- High-performance laptops for each team member
 - Processor: Intel Core i7 or equivalent
 - RAM: 32GB
 - Storage: 1TB SSD
 - Graphics: Dedicated GPU for UI/UX design work

10.2.3 SERVER INFRASTRUCTURE

- Cloud-based servers for application hosting and database management
- Load balancers for traffic distribution and improved performance
- Backup and disaster recovery systems

11 ASSUMPTIONS

- The required APIs from third-party services will be fully operational and accessible throughout the development period.
- The deployment environment will have the necessary storage and processing resources to support project deployment.
- Team members working on the mobile application will have uninterrupted access to testing devices throughout the project cycle.
- All software licenses required for development will be available and valid throughout the project.
- Stakeholders will provide timely feedback during the review and testing phases to avoid delays.

12 CONSTRAINTS

- Final prototype demonstration must be completed by May 1, 2025.
- The system must be fully functional in both the web app and the progressive web app (PWA) versions.
- Development work will be constrained by team availability, with most work expected to occur outside of class hours.
- Total development costs must not exceed \$800.
- All project documentation must be completed and submitted by May 1, 2025.

13 RISKS

14 DOCUMENTATION & REPORTING

14.1 MAJOR DOCUMENTATION DELIVERABLES

14.1.1 PROJECT CHARTER

The Project Charter is the core document of our project, it will describe all key aspects of the project such as schedule, budget, goals and constraints. The charter will be maintained on the Overleaf web

Risk description	Probability	Loss (days)	Exposure (days)
Delays in modernizing the web UI may push the development timeline for the mobile app	0.40	15	6.0
Security vulnerabilities in user authentication and data protection	0.50	20	10.0
Limited team availability slowing down feature development (outside of class hours)	0.30	12	3.6
Delays in integration with Canvas due to API issues or technical difficulties	0.20	14	2.8
Certification delays for security compliance of the mobile app	0.15	10	1.5

Table 1: Overview of highest exposure project risks for LearnLeaf Organizer

application. a joint email account has been created to allow every member access to the charter and other documentation. The initial version of the charter will be delivered on September 23rd 2024 (the end of sprint 1). in order to create and maintain the charter, a task will be created in each sprint and assigned to a different member to review and update the charter as a whole if necessary. The final version will be delivered by December 8th 2024.

14.1.2 SYSTEM REQUIREMENTS SPECIFICATION

The System Requirements Specification is an exhaustive list of the requirements and constraints that our project must meet. It will include information about the general purpose of the project, along with all requirements, ranging from broad system level requirements to fine details such as the order of function handling for the flow of user data. The specification will be maintained on Overleaf and updated biweekly with the other documentation if new requirements are discovered in the development process. Additionally, once per month, the development team will meet and discuss adding or changing requirements to better fit the clients needs. The initial version will be delivered on October 14 2024, and the final version will be delivered December 2, 2024

14.1.3 ARCHITECTURAL DESIGN SPECIFICATION

The Architectural Design Specification is an overview of the system containing detailed descriptions of the systems and subsystems within the project. It will describe the scope and purpose of each of the components as well as the interfaces between each component. The design specification will be maintained in a group account on Overleaf, ensuring access by all members. It will be maintained by a dedicated team member each sprint, and will be updated according to group consensus at each sprint planning meeting. The initial version will be delivered November 2024, and the final version will be delivered in SD 2.

14.1.4 DETAILED DESIGN SPECIFICATION

The Detailed Design Specification is a fine grained explanation of the design features of each subsystem within the project. It will contain diagrams of each subsystems dependencies, along with hardware, OS and programming language information for each subsystem. The design specification will be maintained in a group account on Overleaf, ensuring access by all members. It will be maintained by a dedicated team member each sprint, and will be updated according to group consensus at each sprint planning meeting. The initial version of the detailed design specification will be delivered on December 02, 2024 and the final version will be delivered at the end of SD 2.

14.2 RECURRING SPRINT ITEMS

14.2.1 PRODUCT BACKLOG

Before each sprint, when the development team does the sprint planning, the team will discuss items not yet added to the backlog. We will collectively assign points through averaging each members estimate of value. The priority of each item will be discussed, but will ultimately be the product owners decision. Jira will be used to maintain the backlog and will be available for all team members and the product owner

14.2.2 SPRINT PLANNING

How will each sprint plan be planned? How many sprints will there be (you need to look at the schedules for this course and previous Senior Design II courses during the appropriate semesters to figure this out).

Each sprint will be planned There will be 8 sprints total through senior design 1 and 2.

14.2.3 SPRINT GOAL

The sprint goal will be decided for the team by the scrum master for that sprint based on priority and then demonstration/due date. Our customer will be involved by determining the priority for each epic in jira, thus informing the importance of each feature.

14.2.4 SPRINT BACKLOG

Sprint backlog items will be added based on the sprint goal that has been chosen. The goal or epic(in jira) will have tasks that are assigned to it. during sprint planning, key tasks to accomplish the epic will be assigned first, with the remaining work hours being filled with less important tasks, documentation maintenance, and fixing bugs. The backlog will be maintained in Jira, which will give each team member the ability to mark their tasks in progress, blocked, or finished.

14.2.5 TASK BREAKDOWN

Individual tasks will be assigned through voluntary claiming or assignment by the scrum master. This will attempt to follow individuals skills sets, but a priority will be placed on ensuring that group members regularly receive small tasks outside their normal purview to maintain familiarity with the product as a whole. Time spent will be documented using the "real time spent" feature on Jira, which requires members to input a real number of hours spent on a task before marking it complete.

14.2.6 SPRINT BURN DOWN CHARTS

Burn down charts will be automatically generated by Jira, which tracks hours spent and points by each individual member. Each burndown chart will track story points accomplished, and will graph remaining story points against time.



Figure 2: Example sprint burn down chart

14.2.7 SPRINT RETROSPECTIVE

The sprint retrospective will be created at the post sprint review session. The team will convene and discuss how each member feels the sprint went. Ideas for improving group and individual performance will be documented so that progress can be checked against later sprints. Procedures and methodologies that work well will be identified and continued. It will be due before the beginning of each new sprint

14.2.8 INDIVIDUAL STATUS REPORTS

Individual status reports will be created at the end of each sprint, they will include the individuals estimation of the status of the project. Some key items that will be included are assessment of team

mates performance, and a self reflection on the sprint.

14.3 CLOSEOUT MATERIALS

14.3.1 SYSTEM PROTOTYPE

Because our project is entirely software, there is no need for a system prototype, or a field acceptance test.

14.3.2 PROJECT POSTER

The Project Poster will be 3' by 4', it will include sections on the features of our software, our security measures, and why our software is superior to other organizer software. It will be delivered by the CoE innovation day.

14.3.3 WEB PAGE

The learn leaf organizer is at its core, a web page. This means that the vast majority of the project itself will be available on the project web page. It will be made available to the public from the start in order to smooth the process of beta testing, and to ensure that each update works once it has been deployed. It will naturally be updated as the project progresses and the final delivery will be in May 2025 at the end of SD 2.

14.3.4 DEMO VIDEO

Our demonstration video will cover user registration/login, creation of classes, projects, assignments, manipulation of native calendar events and canvas calendar importing. It will be approximately 15 minutes long. We will not include B reel, but will retain all of that footage for future demonstration purposes.

14.3.5 SOURCE CODE

We will use Github to maintain the source code for our project. Source code will be proprietary, so only binaries will be provided to customers if we implement a full desktop application. Otherwise, we will not need to provide source or binary files as a progressive web application. As a website, we do not need a license, but may implement a privacy policy to meet requirements for ad monetization. If a desktop application is deployed, we will put license terms and information in a ReadMe file.

14.3.6 SOURCE CODE DOCUMENTATION

We will follow the JSDoc3 standard for javascript documentation. We currently plan to manually create documentation for any proprietary parts of our code base, providing links for existing documentation for imported tools such as firebase. Final documentation will be provided in in HTML format.

14.3.7 INSTALLATION SCRIPTS

Deployment of our software will be very simple. There will be no need for an executable download, as we will deploy our software as a progressive web application. This will allow us to install our web application without an executable on a PC and without needing to meet requirements to put our mobile application on the android or apple app store.

14.3.8 USER MANUAL

The software should be self explanatory and intuitive. We may include tool tips and other helpful navigation features, but there is no plan for a user manual. If the project becomes more complex than anticipated, a short walk through may be programmed to initiate on first login.

REFERENCES